which the sample is maximized. As we can see, one sample is maximized around -3.2, another around -1.2, and the other at 5.

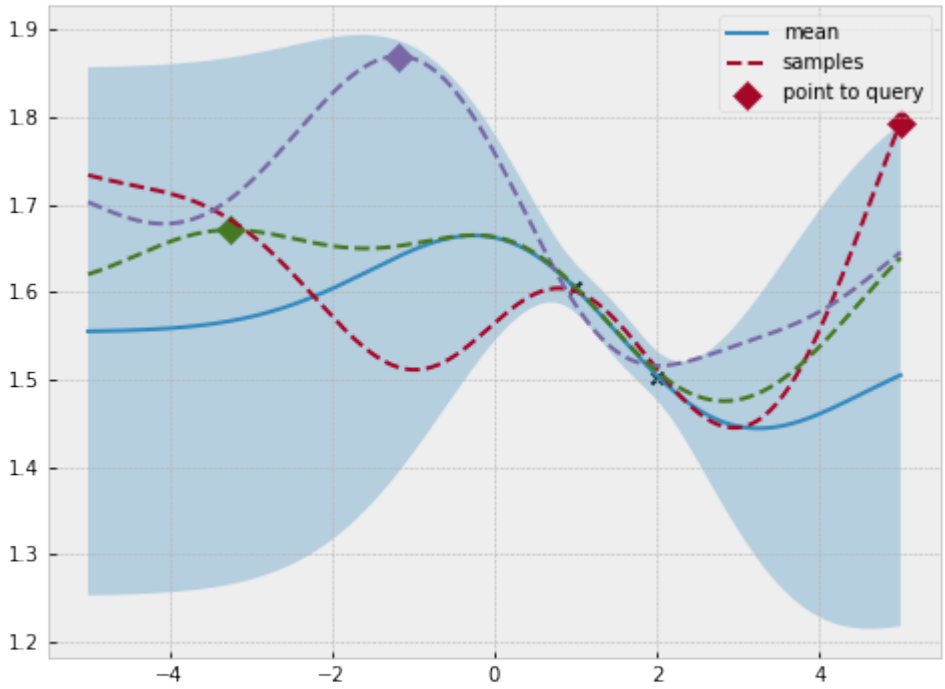


Figure 5.12 Samples drawn from a Gaussian process and data points that maximize the corresponding samples. Whichever sample is drawn, Thompson sampling selects the data point that maximizes that sample as the next point to evaluate the objective function at.

When using the TS policy, it's entirely up to chance which of the three samples above (or a complete different sample) we draw from the GP. However, whichever sample we draw, the data point that maximizes the sample is the one that we will query next. That is, if we draw the sample maximized at -3.2, then -3.2 is where we will evaluate the objective function next. If we draw the sample maximized at 5, then we will query the point $x = 5$ next.

IMPORTANT **Acquisition scores of Thompson sampling**

The acquisition scores that Thompson sampling computes is the values of a random sample drawn from the Gaussian process. The data point that maximizes this sample is the one that we will query next.

Unlike other policies we have seen so far, Thompson sampling is a *randomized* policy, which means that when faced with the same training data and Gaussian process, it's not guaranteed that the policy will make the same decision. However, this randomness is in no way a disadvantage. We have said that drawing a sample from a GP is easy, so computing the TS acquisition score may be done efficiently. Further, maximization of a random sample inherently balances between exploration and exploitation, which as we know is the main concern in BayesOpt.

- If a data point has a high predictive mean, then the value of a random sample at that data

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